

DÚ č. 1

1.1) Dokážte, že π ľubovoľným komplexným VP V so skalárnym súčinom

$$\langle, \rangle \text{ platí } \operatorname{Re}(\langle w, \pi \rangle) = \frac{1}{4} (\|w + \pi\|^2 - \|w - \pi\|^2)$$

$$\operatorname{Im}(\langle w, \pi \rangle) = \frac{1}{4} (\|w - i\pi\|^2 - \|w + i\pi\|^2)$$

DK

$$\rightarrow \text{platí } \|x\|^2 = \langle x, x \rangle \quad \langle w, \pi \rangle = \overline{\langle \pi, w \rangle}$$

Reálna časť

$$\begin{aligned} 1) \|w + \pi\|^2 &= \langle w + \pi, w + \pi \rangle \\ &= \langle w, w \rangle + \langle w, \pi \rangle + \langle \pi, w \rangle + \langle \pi, \pi \rangle \end{aligned}$$

$$2) \|w - \pi\|^2 = \langle w, w \rangle - \langle w, \pi \rangle - \langle \pi, w \rangle + \langle \pi, \pi \rangle$$

$$\begin{aligned} \rightarrow \|w + \pi\|^2 - \|w - \pi\|^2 &= \langle w, w \rangle + \langle w, \pi \rangle + \langle \pi, w \rangle + \cancel{\langle \pi, \pi \rangle} - (\langle w, w \rangle - \langle w, \pi \rangle - \langle \pi, w \rangle + \cancel{\langle \pi, \pi \rangle}) \\ &= 2\langle w, \pi \rangle + 2\overline{\langle w, \pi \rangle} \end{aligned}$$

$$\rightarrow \|w + \pi\|^2 - \|w - \pi\|^2 = 2\langle w, \pi \rangle + 2\overline{\langle w, \pi \rangle} = 4\operatorname{Re}\langle w, \pi \rangle$$

$$\rightarrow \operatorname{Re}\langle w, \pi \rangle = \frac{1}{4} (\|w + \pi\|^2 - \|w - \pi\|^2) \quad \checkmark$$

Imaginárna časť

$$\begin{aligned} 1) \|w + i\pi\|^2 &= \langle w + i\pi, w + i\pi \rangle \\ &= \langle w, w \rangle + i\langle w, \pi \rangle - i\langle \pi, w \rangle + \langle \pi, \pi \rangle \end{aligned}$$

$$2) \|w - i\pi\|^2 = \langle w, w \rangle - i\langle w, \pi \rangle + i\langle \pi, w \rangle + \langle \pi, \pi \rangle$$

$$\begin{aligned} \rightarrow \|w - i\pi\|^2 - \|w + i\pi\|^2 &= \langle w, w \rangle - i\langle w, \pi \rangle + i\langle \pi, w \rangle + \cancel{\langle \pi, \pi \rangle} - (\langle w, w \rangle + i\langle w, \pi \rangle - i\langle \pi, w \rangle + \cancel{\langle \pi, \pi \rangle}) \\ &= -2i\langle w, \pi \rangle + 2i\langle \pi, w \rangle \end{aligned}$$

$$\rightarrow \langle \pi, w \rangle = \overline{\langle w, \pi \rangle} \quad \rightarrow$$

$$\rightarrow = 2i(\overline{\langle w, \pi \rangle} - \langle w, \pi \rangle)$$

$$\boxed{\overline{z} - z = -2i\operatorname{Im}(z)} \quad z = \langle w, \pi \rangle$$

$$\rightarrow \|w - i\pi\|^2 - \|w + i\pi\|^2 = 4\operatorname{Im}\langle w, \pi \rangle$$

$$\rightarrow \operatorname{Im}\langle w, \pi \rangle = \frac{1}{4} (\|w - i\pi\|^2 - \|w + i\pi\|^2) \quad \checkmark$$